```
// PA1301: PERFORM LARGE CALCULATION

const startBtn = document.getElementById("startBtn");
const statusText = document.getElementById("status");
const progressText = document.getElementById("progress");
const resultText = document.getElementById("result");

let worker = null;

// PA1301: Start calculation
startBtn.addEventListener("click", function() {

    // Stop previous Worker if one exists
    if (worker !== null) {
        worker.terminate();
    }

    // Create Web Worker
    const currentWorker = new Worker("workers.js");
    worker = currentWorker;

    statusText.textContent = "Calculating...";
    progressText.textContent = "Progress: 0%";
    resultText.textContent = "Result: Please wait...";

    // PA1301: Receive messages from Worker
    currentWorker.onmessage = function(event) {

        if (worker !== currentWorker) {
            return;
        }

        const data = event.data;

        // Handle progress updates
        if (data.type === "progress") {

            progressText.textContent =
                "Progress: " + data.result + "%";

            console.log("Progress:", data.result + "%");
```

workers.js > onmessage

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//pa1301


self.onmessage=function(event){
    const limit=event.data
    let total =0;
    let lastprogress = 0;

    for (let i = 1; i <=limit; i++){
    sum+=i;
if(i %1000000 ===0){
    const percentage=Math.round(
       ( i/limit ) *100
    );

    self.postMessage({
        type:"progress",
        result: percentage,
    });
}
    }
    self.postMessage({
        type:"result",
        result:total
```

```
<button id="startBtn">Start Calculation</button>
<script src="./app.js"></script>
```

large calculation

Start Calculation test

result